B-Energy Conversion MS4092

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Number of problems: 8
Start at: 10.30 AM
Finish at: 10.53 AM
Duration: 23 Minutes

The relative humidity is defined as

- ratio of (saturation pressure of pure water) to (ambient pressure)
- ratio of (ambient pressure) to (saturation pressure of pure water)
- ratio of (partial pressure of water vapor) to (saturation pressure of pure water)
- ratio of (saturation pressure of pure water) to (partial pressure of water vapor)

Please answer the following problem

The view factors of an enclosure with four surfaces are as follow,

$F_{11} = 0.2$

$F_{12} = 0.3$

$F_{13} = 0.25$

$F_{11} + F_{12} + F_{13} + F_{14} = 1$

$F_{14} = 0,25$

If $A_1 = 3\ m^2$ and $A_4 = 2\ m^2$, determine the view factor of F_{41}

- 0.375
- 0.50
- 0.125
- 0.25

$F_{ij} \cdot A_i = F_{ji} \cdot A_j$

$F_{41} = F_{14} \cdot \frac{A_1}{A_4}$

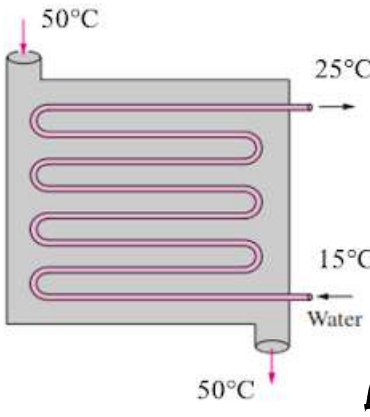
$= 0,25 \cdot \frac{3}{2}$

$F_{41} = 0,375$

Please answer the following problem

A condenser of a steam power plant has a condensing steam at the temperature of 50°C. The cooling water (C_p = 4200 J/kg·°C) enters the tubes of the condenser at 15°C and leaves at 25°C. The surface area of the tubes is 50 m², and the overall heat transfer coefficient is 2500 W/m² °C. Determine the mass flow rate of the cooling water required.

$U = 2500\ W/m^2$



$T_h = 50°C$

$T_{c,1} = 15°C$

$T_{c,2} = 25°C$

$C_p = 4200\ J/kg°C$

$A: 50\ m^2$

$\Delta T_1 = T_{h,1} - T_{c,2} = 25°C$

$\Delta T_2 = T_{h,2} - T_{c,1} = 35°C$

$$\Delta T_m = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)} = 29.72°C$$

$\dot{Q} = U \cdot A \cdot \Delta T_m = 3715000\ W$

- ○ 123.83 kg/s
- ✗ 88.45 kg/s
- ○ 84.91 kg/s
- ○ 106.14 kg/s

$$\dot{m} = \frac{\dot{Q}}{C_p \cdot \Delta T_c} = 88.45\ kg/s$$

A series configuration of two Carnot engines is designed so that the heat rejected from the first engine is utilized as heat input for the second engine. The temperature limits are 1050K and 300K for the hot and cold reservoirs, respectively. If the ratio of the work produced by engine 1 : engine 2 is 1:2, what is the intermediate temperature between the engines?

$T_h = 1050\ K \quad T_c = 300\ K$

- ✗ 800 K
- ○ 1000 K
- ○ 900 K
- ○ 700 K

$W_2 = 2W_1$

$T - T_c = 2(T_h - T)$

$T - 300 = 2(1050 - T)$

$3T = 2400$

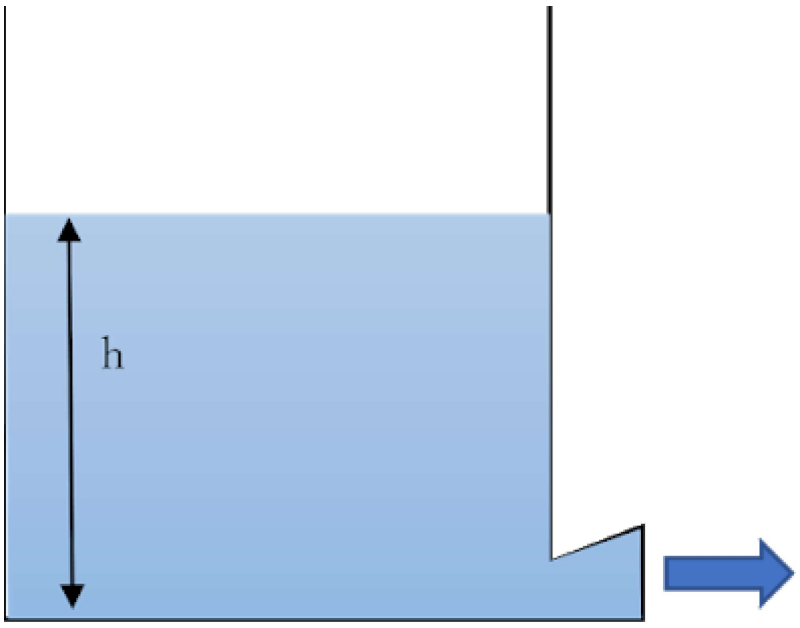
$T = 800\ K$

Please answer the following problem

A new hall in Russia is being designed hall with a capacity of 5000 people. In order to prevent Covid-19 infection, the recirculation flow is deactivated with the ventilation rate from fresh air of 4 x 10^5 m^3/h. The ambient pressure and temperature are 100 kPa and -10°C, respectively. If the sensible heat from the attendants is 5 MW and air is assumed as dry air with the specific heat is 1.0048 kJ/kg.K, what is the temperature of the hall?

- ○ 12.5°C
- ○ 17°C
- ○ 35.1°C
- ○ 23.8°C

A water tank filled with water up to a height of h. Water is discharged through a small nozzle shown in the following figure. What kind of flow is in the pipe before the nozzle?



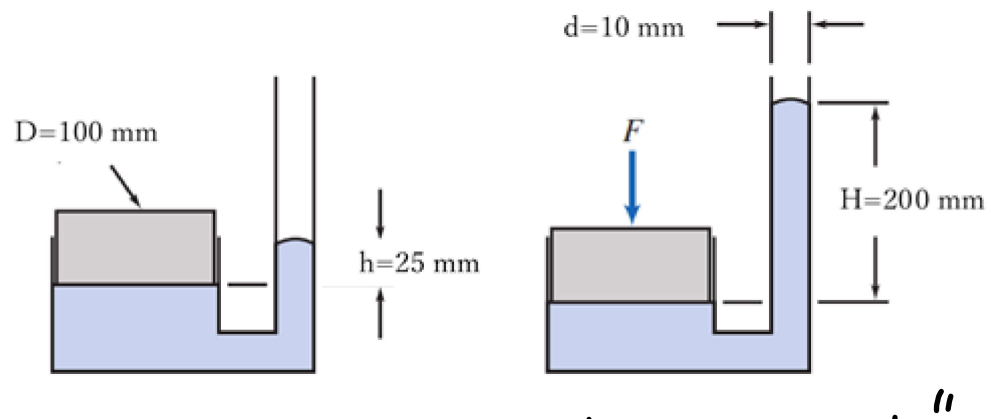
- [x] steady and non-uniform
- [ ] unsteady and non-uniform
- [ ] steady and uniform
- [ ] unsteady and uniform

A hot boiled egg is placed in a spacecraft that is filled with air at atmospheric pressure and temperature. How the egg temperature decreases when the spacecraft is in space compared to when it is on the ground?

- [ ] Temperature does not decrease
- [ ] Slower
- [ ] Decrease at the same rate
- [x] Faster

gaada konveksi bebas
↳ gaada perpindahan panas
temperatur turun

The tube shown is filled with water (ρ=1000kg/m^3). Calculate the force applied to the piston. (g = 9.81 m/s^2)



- [ ] 17.17 N
- [ ] 31.88 N
- [ ] 22.07 N
- [ ] 26.98 N

"tekanan sama ke segala arah"

$P = \rho \cdot g \cdot \Delta h = 1000 \cdot 9{,}81 \cdot \frac{(200-25)}{1000} = 1716{,}75\ Pa$

$F = P \cdot A = 1716{,}75 \times \frac{\pi}{4} \cdot 0{,}1^2 = 13{,}48\ N$

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